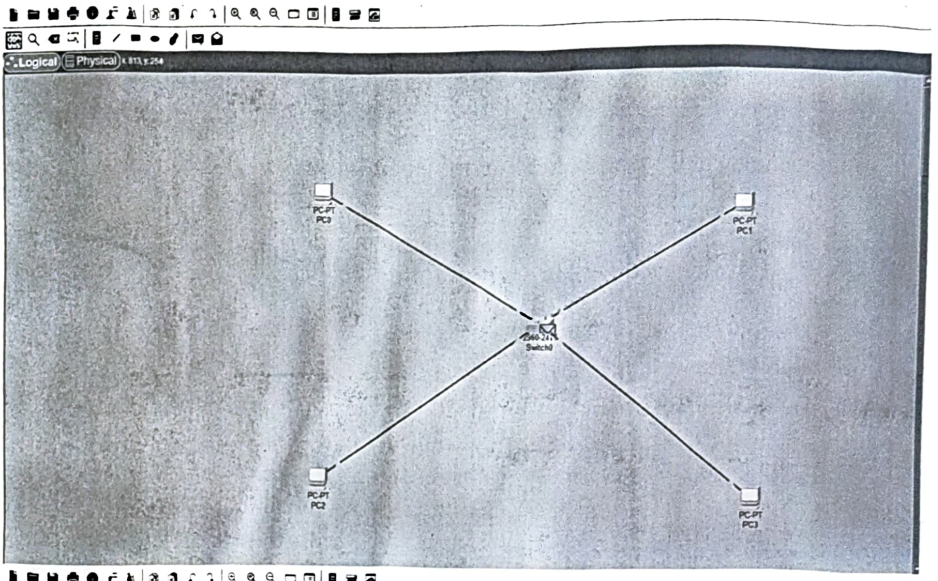
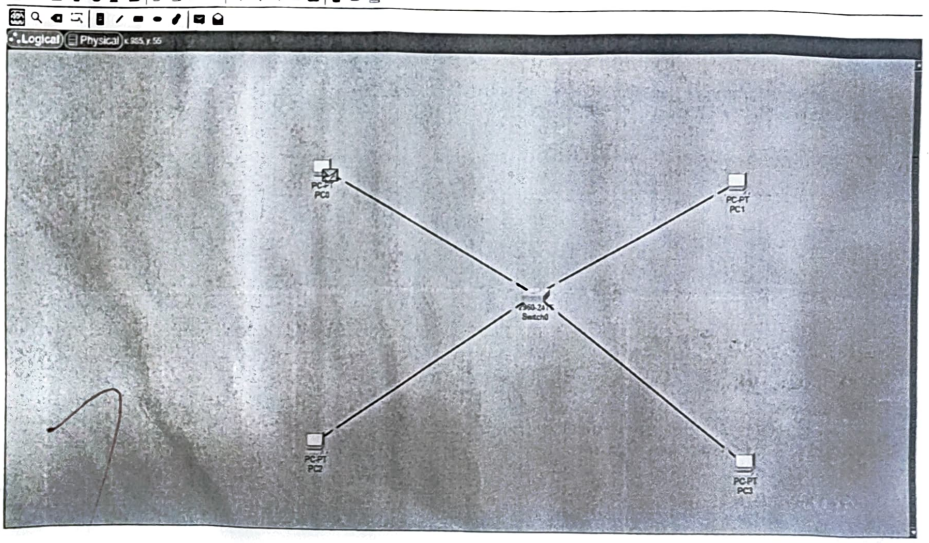




Result:

Thus to Design a simple network with multiple nodes and connect via networking devices available in library is implemented successfully.

Experiment-2

DIFFERT TYPES OF NETWORK CABLE

a) Understand different types of cables used in networking are:-

1) Unshielded Twisted Pair (UTP)
2) Shielded Twisted Pair (STP)
3) Coaxial
4) Fibre optic cable.

UTP:-

:- category	Max data Transmission	Use
Category 3 -	10bps	10Base-T - Ethernet
category 5	100Mbps	Fast Ethernet, Gigabit Ethernet
Category 5e	1Gbps	Fast Ethernet, Gigabit Ethernet

STP:-

Category 6/6a	10Gbps	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centers

SSTP :-

Category 7	10Gbps	Gigabit Ethernet, 10G Ethernet (100m)

Coaxial cable :-

RG-6 RG-59 RG-11	10-100Mbps	Speed of signal is 5cm Television network High speed Internet connection.

fiber optics cable

Single mode, Multi mode,	100Gbps	Maximum distance of fiber optics cable is around 100 meters

b) Make your own Ethernet cross - over cable / Straight cable.

Tools and Parts Required:-

* CAT5/CAT5e Ethernet cable
* Crimping tool
* Two RJ45 connectors
* Plug shields

Step 1:- To start construction of the device, begin by threading shields onto the cable

Step 2:- Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:-

Untwist the four wire pairs and arrange them according to A and B

Step 4:-

Straighten and trim the wires evenly, then present them into the RJ45 connector.

Step 5:-

Push the wires fully into the RJ45 plug.

Step 6:-

Insert the plug into the crimping tool & crimp it

Step 7:-

Repeat the same process on the other end using B for crossover and A for straight-through cable.

Result:-

Thus the ~~study of~~ different types of network cabel used in networking are studied.

